



# Chapter 8 Monte Carlo Methods

# Task 1.1 Use ancestral sampling to generate $S=1000$ samples from joint distribution $p(x_1, x_2, x_3, x_4)$

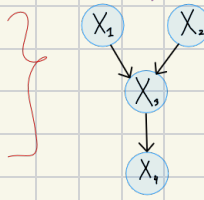
- we have:

$$x_1 \sim \mathcal{N}(0, 3)$$

$$x_2 \sim \mathcal{N}(1, 1)$$

$$x_3 | x_2, x_1 \sim \mathcal{N}(x_1 + x_2, 2)$$

$$x_4 | x_3 \sim \mathcal{N}(x_3^2, 1)$$



- we use the product rule  $P(A, B) = P(B|A) P(A)$

$$p(x_4 | x_3, x_2, x_1) p(x_3 | x_2, x_1) p(x_2 | x_1) p(x_1)$$

○ we can see from the model that  $x_4$  depends only on  $x_3$ ,  $\therefore$  we use a conditional independence  $x_4 \perp \{x_1, x_2\} | x_3$  so it reduces to  $P(x_4 | x_3)$

○  $x_3$  depends on  $x_2, x_1$  so this term remains  $P(x_3 | x_2, x_1)$

○ From the model graph we see, that  $x_2 \perp x_1$ , so we can reduce  $P(x_2 | x_1) P(x_1)$  to  $P(x_2) P(x_1)$

$\Rightarrow$  Putting it all together:  $P(x_4 | x_3) P(x_3 | x_2, x_1) P(x_2) P(x_1)$

If we have following hierarchical model:  $x_1 \rightarrow x_2 \rightarrow x_3 \rightarrow x_4$

would decompose to  $p(x_4 | x_3) p(x_3 | x_2) p(x_2 | x_1) p(x_1)$

## Monte Carlo sampling

○ our target distribution  $\mathcal{N}(1, 5)$

○ train params = 1

○  $\epsilon_{\text{tol}} = 2.0$

○  $\text{theta\_init} = 0$

○  $\text{theta\_l1\_fee} = 0$

○  $\text{log\_target} = \text{lambda } x$

○  $\text{theta\_cur} = \text{theta}[-1]$

$\log\text{-pdf}(x, 1, 3)$

as initial, we have  $\text{theta\_init} = 0$

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- plugging in  $\mu=1$  &  $\sigma^2=3$ , how we take a log:  $\log p(x) = \ln\left(\frac{1}{\sqrt{2\pi \cdot 3}}\right) \exp\left(-\frac{(x-1)^2}{2 \cdot 3}\right)$

$$= \ln(1) - \ln\left[\left(\frac{6\pi}{1}\right)^{\frac{1}{2}}\right] - \frac{1}{6}(x-1)^2$$

$$= -\frac{1}{2} \ln(6\pi) - \frac{1}{6}(x-1)^2$$

$$\log p(x) = -\frac{1}{6}(x-1)^2 + C$$

$\leftarrow$  we will use the following to evaluate our Proposal

○ how we generate a new proposal:  $\text{theta\_stor} = \text{theta\_cur} + \epsilon_{\text{tol}} \cdot \text{normal}(0, 2)$

○ this is a random walk w/  $\gamma = 2$ , i.e.  $\theta^{k+1} \in \mathbb{R}^n$ ,  $\epsilon \sim \mathcal{N}(0, \gamma)$

## Iteration 1 ( $k=1$ )

○ 1. Current state

$\Leftarrow$  code:  $\text{theta\_cur} = \text{theta}[-1] \Rightarrow x^0 = 0.0$

$\Leftarrow$  code:  $\log\text{-p\_theta} = \log\text{-target}(\text{theta\_init}) \Rightarrow \log p_e(x) = C - \frac{(0-1)^2}{6} = C - \frac{1}{6} = -1.16 - 0.16 = -1.32$

○ 2. Propose

$\Leftarrow$  code:  $\text{theta\_stor} = \text{theta\_cur} + \epsilon_{\text{tol}} \cdot \text{normal}(0, 2) = x^* = 0.0 + 1.20 = 1.20 = x^* = \text{theta\_stor}$   
 $\epsilon_{\text{tol}} = 1.20$  i.e. the value generated from  $\mathcal{N}(0, 2)$   
 $0$  is the initial value

○ 3. Evaluate target at proposal

$\Leftarrow$  code:  $\log\text{-p\_theta\_stor} = \log\text{-target}(\text{theta\_stor}) \Rightarrow \log p_e(1.20) = C - \frac{(1.2-1)^2}{6} = C - \frac{0.04}{6} = -1.16 - 0.01 = -1.17 = 0^* = \log\text{-p\_theta\_stor}$

○ 4. Acceptance ratio

$$= \log p_e(0^*) - \log p_e(0^{\text{cur}})$$

$\Leftarrow$  code:  $\log\text{-r} = \log\text{-p\_theta\_stor} - \log\text{-p\_theta} \Rightarrow \log r = -1.17 - (-1.32) = 0.16$

$\bullet \log\text{-r} > 0 \Rightarrow$  the candidate is at higher density  $\Rightarrow$  should be easier to move there. Candidate has higher density and  $A=1$ . We always accept w/ prob 1.0

$\bullet \log\text{-r} < 0 \Rightarrow$  the candidate is at lower density  $\Rightarrow$  move only sometimes.  $A = \exp(\log\text{-r}) \in (0, 1)$ . We accept w/ prob  $A$ , else reject.

## 5. Acceptance prob

code:  $A = \min(1, \exp(\log-r)) \Rightarrow A_1 = \min(1, e^{\log-r}) = \min(1, e^{0.16}) = \min(1, 1.17) = 1$

- we calculate in  $\log$  for numerical stability, then  $\exp$  to get probabilities

## 6. Accept/Reject

code: if  $\text{random.uniform}(\text{key\_accept}) < A \Rightarrow$  in the uniform draw we get  $u_1 = 0.70 \therefore u_1 = 0.70 < A = 1 \Rightarrow \text{Accept}$

code:  $\text{set } \theta_{\text{next}} = \theta_{\text{cur}}$ ,  $\text{update } \log\_p\_theta = \log\_p\_theta_{\text{cur}}$ ,  $\text{append } \text{accepts.append}(1)$

this is just a Bernoulli coin flip w/ success prob  $A$

- the algo must randomize w/ prob  $A$  to make the trans kernel to satisfy detailed balance w.r.t. the target. This condition guarantees the chain's stationary distribution is  $P_\pi$ . If we always accept only "uphill" and always reject "downhill", the chain would greedily climb to a mode and get stuck and not sample the distribution.

- if  $u < A \Rightarrow \text{accept} : \theta_{\text{next}} = \theta_{\text{cur}}$ , and replace  $\log\_p\_theta$  w/  $\log\_p\_theta_{\text{cur}}$  (we moved, so the old log-density must follow the state)

- else:  $\Rightarrow$  Reject.  $\theta_{\text{next}} = \theta_{\text{cur}}$  and keep  $\log\_p\_theta$  (state didn't change)

what keeps the chain from getting stuck and ensures correct long run sampling.

## 7. State update

code:  $\theta_{\text{nexts.append}}(\theta_{\text{next}}) \Rightarrow x^* = 1.20$

## Iteration 2 ( $k=2$ )

### 1. Current state

code:  $\theta_{\text{cur}} = 1.20$

code:  $\log\_p\_theta = -1.47$

### 2. Propose

code: Add  $\epsilon_2 = -3.50 : x^* = 1.20 - 3.50 = -2.30$

### 3. Evaluate target at proposal

code:  $\log\_p\_theta_{\text{cur}} = \log_\pi(-2.30) = C - \frac{(-2.3-1)^2}{6} - C - \frac{(-3)^2}{6} = C - \frac{10.89}{6} = -1.46 - 1.815 = -3.28$

### 4. Acceptance ratio

$$\log r_2 = -3.28 - (-1.47) = -1.80$$

### 5. Acceptance prob

$$A_2 = \min(1, e^{-1.80}) = 0.16$$

### 6. Accept/Reject

$u_2 = 0.40$ , compare  $0.40 < 0.16 \Rightarrow \text{no} \Rightarrow \text{reject}$

code path:  $\theta_{\text{next}} = \theta_{\text{cur}}$ ,  $\text{accepts.append}(0)$

### 7. State update

Append  $x^* = 1.20$  (unchanged due to rejection)

Task 23 what is the resulting accept. prob.  $A_k$  when the candidate sample  $\theta^*$  has higher density value than the previous value  $\theta^{k-1}$ ?

○  $P_k(\theta^*)$ : the current state of the Markov chain b4 you make the  $k$ -th update. It's where the chain is sitting now.

○  $P_k(\theta)$ : The candidate (proposal) drawn given the current state:  $\theta^* \sim q(\cdot | \theta^{k-1})$

□ After the decision you get the next state:

$$\theta^k = \begin{cases} \theta^* & \text{with prob. } A_k = \min(1, r_k) \\ \theta^{k-1} & \text{otherwise reject} \end{cases}$$

○ ex. at iteration  $k$ :

• current  $\theta^{k-1} = 1.2 \Rightarrow \log P_k(\theta^{k-1}) = -1.42$

• Proposal  $\theta^* = 2.1 \Rightarrow \log P_k(\theta^*) = -1.66$

• then  $\log r = -1.66 - (-1.42) = -0.24$ ;  $A = \min(1, e^{\log r}) = \min(1, e^{-0.24})$

• Draw  $u \sim U(0,1)$ . If  $u < 0.783$ , accept and set  $\theta^k = 2.1$   
Else reject and keep  $\theta^k = 1.2$

Intuition: Random walks.  $\theta^{k-1}$  is where the walker stands now.  $\theta$  is a proposed step. The acceptance rule decides whether to take the step or stay put, so that over time, the walker spends time in regions proportional to the target density  $p_k$ .

If the candidate  $\theta^*$  has higher density than the current state  $\theta^{k-1}$ , then  $A_k = \min\{1, r_k\} = 1$ , so we accept w/ probability 1.

So if Candidate has higher density  $P_k(\theta^*) > P_k(\theta^{k-1}) \Rightarrow r_k > 1 \Rightarrow A_k = \min(1, r_k) = 1$ .

Task 24 Use  $r_k = \frac{P_k(\theta^*) q(\theta^{k-1} | \theta^*)}{P_k(\theta^{k-1}) q(\theta^* | \theta^{k-1})}$  to argue that we don't need to be able to evaluate the target distribution, but only the target density of up to a constant. That is, we only need to evaluate  $P_k$ . Why is it important in modelling?

MH acceptance ratio  $r_k = \frac{P_k(\theta^*) q(\theta^{k-1} | \theta^*)}{P_k(\theta^{k-1}) q(\theta^* | \theta^{k-1})}$ ,  $A_k = \min(1, r_k)$

Spec. our target posterior is  $P_k(\theta) = P(\theta | y) = \frac{P(y | \theta) P(\theta)}{P(y)}$ , plug this into  $r_k$

$$\frac{\frac{P(y | \theta^*) P(\theta^*)}{P(y)} q(\theta^{k-1} | \theta^*)}{\frac{P(y | \theta^{k-1}) P(\theta^{k-1})}{P(y)} q(\theta^* | \theta^{k-1})} = \frac{P(y | \theta^*) P(\theta^*) q(\theta^{k-1} | \theta^*)}{P(y | \theta^{k-1}) P(\theta^{k-1}) q(\theta^* | \theta^{k-1})}$$

! This is important bcs many models  $p(y) = \int P(y, \theta) = \int P(y | \theta) p(\theta) d\theta$  is high-dimensional, intractable integral. MCMC allows us to sample from  $P_k(\theta) \propto p(y | \theta) p(\theta)$ , w/out needing to compute  $p(y)$ .

Task 3.5 Use MII-algo to estimate mean  $E[\theta]$ , variance  $V[\theta]$  and probability that  $p(\theta > 1)$ .

! We must provide log of the target density !

◦  $p(\theta) \propto \exp(-|\theta-3|)$

- we see that our distribution looks like Laplace Distribution :

$$p(\theta) = \frac{1}{2b} \exp\left(-\frac{|\theta-\mu|}{b}\right)$$

- we observe that  $b=1$  and  $\therefore p(\theta) = \frac{1}{2} \exp(-|\theta-3|)$

- taking log, we obtain  $\log p(\theta) = -\log(2) - |\theta-3|$

# Poisson Distribution Bayes + MCMC

## Step 1: Standardize age

$$x_n = \frac{\text{age}_n - \overset{\text{mean}}{m_{\text{age}}}}{\underset{\text{std}}{s_{\text{age}}}}, \quad m_{\text{age}} = \frac{35+64}{2} = 49.5; \quad s_{\text{age}} \approx 8.65 \text{ (for age 35 to 64)}$$

## Dataset prep

- $(\text{age}, y) = (35, 3), (36, 1), (37, 3)$
- standardized =  $(-1.676, -1.560, -1.445)$
- design vectors  $x_n = [1, x_n]^T$

## Model

- likelihood for each  $n$ :

$$y_n | \mu_n \sim \text{Poisson}(\mu_n), \quad \text{PMF} \quad P(y_n = y | w, a_n) = \frac{\mu(a_n)^y e^{-\mu(a_n)}}{y!}, \quad y = 0, 1, 2, \dots$$

- mean and variance of Poisson:

$$E[y_n | w, a_n] = \mu(a_n) \quad \text{Var}[y_n | w, a_n] = \mu(a_n)$$

- log-likelihood

o for single data pt. at age  $a_n$

$$\log p(y_n | w, a_n) = y_n f_n - \exp(f_n) - \log(y_n!), \quad f_n = w_0 + w_1 x_n = w_0 + w_1 \left( \frac{a_n - m_{\text{age}}}{s_{\text{age}}} \right)$$

o all data  $\sum_{n=1}^N \log p(y_n | w, a_n)$  (constants can be dropped for opt/MH routines)

$$\log L(w) = \sum_{n=1}^N (y_n f_n - e^{f_n}) - \sum_{n=1}^N \log(y_n!)$$

$$\mu_n = \exp(f_n), \quad f_n = w^T x_n = w_0 + w_1 x_n$$

$$\mu(a_n) = \exp(w_0 + w_1 x_n)$$

- prior + hyperprior

$$w_0, w_1 \sim \mathcal{N}(0, \frac{1}{2} I_2) \quad b \sim \mathcal{N}_+(0, 1)$$

## Log prior

$$\log p(w | b) = -\frac{1}{2b} \|w\|^2 - \frac{1}{2} \log(2\pi b), \quad \log p(b) = -\frac{1}{2} \log \frac{b}{2} - \frac{b}{2} \quad (b > 0)$$

# 0 Mh sample

- we'll sample  $\theta = [w_0, w_1, b]$

-  $w_0 = 0.85, w_1 = 0.20, b = 1$

o compact linear predictor  $f_n$ , means  $\mu_n = \exp(f_n)$  and log. lik

- we use  $f_n = w_0 + w_1 x_n$

$$x_1 = -1.676$$

$$f_1 = 0.85 + 0.20(-1.676) = 0.5148, \Rightarrow \mu_1 = \exp(0.5148) \approx \underline{1.673}$$

$$x_2 = -1.560$$

$$f_2 = 0.85 + 0.20(-1.560) = 0.5380, \Rightarrow \mu_2 = \exp(0.5380) \approx 1.713$$

$$x_3 = -1.445$$

$$f_3 = 0.85 + 0.20(-1.445) = 0.5610 \Rightarrow \mu_3 = \exp(0.5610) \approx 1.753$$

Now

$$\begin{aligned} \log L(w) &= \sum_{n=1}^3 (y_n f_n - \exp(f_n)) - \log(3!) - \log(1!) - \log(3!) \\ &= (3 \cdot 0.5148 - 1.673) - (1 \cdot 0.5380 - 1.713) + (3 \cdot 0.5610 - 1.753) - (1.794 + 0 + 1.794) \\ &= (-1.9202) - (-1.1749) - (-1.864) \approx \underline{-4.9521} \end{aligned}$$

Log-prior terms at  $k=1$ :

$$\bullet \|w\|^2 = 0.85^2 + 0.20^2 = \underline{0.7625}$$

$$\bullet \log p(w|1) = -\frac{1}{2} \cdot 0.7625 - \log(2\pi) \approx -0.3813 - 1.10329 \approx -2.2142$$

$$\bullet \log p(b=1) = \frac{1}{2} \log\left(\frac{2}{\pi}\right) - \frac{1}{2} \approx -0.2253 + 0.5 \approx -0.7253$$

$$\therefore \text{the log-joint is } \log p(y, w, b) = \log L + \log p(w|b) + \log p(b) \approx -4.9521 - 2.2142 - 0.7253 \approx -7.9021$$

o Pick  $age^* = 50$ . Standardize:  $x^* = \frac{50 - 49.5}{8.65} \approx 0.008, x^* = [1, x^*]^T$

for a new input  $x^* = [1, x^*]^T$ , the post. pred. dist. integrates out the uncertainty in  $w, b$ :

$$p(y^* | y, x^*) = \iint p(y^* | w, x^*) p(w, b | y) dw db$$

- we can't do the integral in closed form, so we approximate it w/ MH draws  $\sum_{i=1}^S w^{(i)}, b^{(i)} \sum_{i=1}^S$

- for each draw  $i$ :

$$f_*^{(i)} = w^{(i)} x^*$$

$$\mu_*^{(i)} = \exp(f_*^{(i)})$$

$$y_*^{(i)} = \text{Poisson}(\mu_*^{(i)})$$

$$\Rightarrow \text{then pred. Mean is } E[y^* | y, x^*] \approx \frac{1}{S} \sum_{i=1}^S \mu_*^{(i)}$$

$$\text{pred. variance is } V(y^* | y, x^*) \approx \underbrace{\frac{1}{S} \sum_{i=1}^S \mu_*^{(i)}}_{E[\mu_*]} + \underbrace{\frac{1}{S} \sum_{i=1}^S (\mu_*^{(i)} - \bar{\mu}_*)^2}_{V(\mu_*)}$$

$$\bullet \text{pred. pmf } \forall y: p(y^* = y | y, x^*) \approx \frac{1}{S} \sum_{i=1}^S \frac{(\mu_*^{(i)})^y e^{-\mu_*^{(i)}}}{y!}$$

o One accepted MH draw  $\theta = (w_0, w_1) = (0.95, 0.05)$

$$f^* = w^T x^* = w_0 + w_1 x^* = 0.95 + 0.05 \cdot 0.008 \approx 0.9529$$

$$\mu^* = \exp(f^*) \approx \exp(0.9529) \approx \underline{2.593}$$



The poisson pmf at that single draw is:

$$P(y^*=y | \mu^*=2.593) = \frac{2.593^y e^{-2.593}}{y!}$$

- we draw 3 samples:  $w^{(1)} = (0.95, 0.05)$ ,  $w^{(2)} = (0.80, 0.10)$ ,  $w^{(3)} = (1.10, 0.00)$

Then

$$f_*^{(1)} = w_* + w_{*'} x^* = 0.95 + 0.05 \cdot 0.059 = 0.9529 \quad \Rightarrow \mu_*^{(1)} = e^{0.9529} \approx 2.593$$

$$f_*^{(2)} = 0.80 + 0.10 \cdot 0.058 = 0.8058 \quad \Rightarrow \mu_*^{(2)} = e^{0.8058} \approx 2.240$$

$$f_*^{(3)} = 1.10 + 0.0580 = 1.10 \quad \Rightarrow \mu_*^{(3)} = e^{1.10} \approx 3.004$$

Predictive mean (MC)  $\Rightarrow \bar{\mu}_* = \frac{2.593 + 2.240 + 3.004}{3} \approx 2.612$

Pred. var.

- $\text{Var}(\mu_*) \approx \frac{1}{3} [(2.593 - 2.612)^2 + (2.240 - 2.612)^2 + (3.004 - 2.612)^2] \approx 0.0975$
- $\text{Var}(y^*) = \bar{\mu}_* + \text{Var}(\mu_*) = 2.615 + 0.0975 \approx \underline{2.7095}$

Pred. pmf. is mixture of 3 poissions.

$$P(y^*=0) = \frac{1}{3} [e^{-2.593} + e^{-2.240} + e^{-3.004}] \approx \frac{1}{3} (0.0749 + 0.1064 + 0.0496) \approx 0.077$$

$$P(y^*=1) = \frac{1}{3} [2.593 e^{-2.593} + 2.240 e^{-2.240} + 3.004 e^{-3.004}] \approx \underline{0.194}$$

prob next obs. will be exactly 0.